Trainings

General Information

The use of quality engine lubricating oils and appropriate oil drain and filter change intervals are critical factors in maintaining engine performance and durability.

Cummins Inc. recommends the use of oil that meets the American Petroleum Institute (API) performance categories of CF-4, CG-4, CF-4/SG, or CG-4/SH. Oil with an older API classification of CD or CE can be used in areas of the world outside North America where oils meeting the current API categories are **not** available. However, if using CD or CE classification oil, the oil **must** be changed at the standard service interval and **only** extended if scheduled oil sampling is used for close monitoring of oil condition. Oil with an API classification of CC can be used in areas of the world outside North America where oils meeting the current API categories are **not** available. If used, they **must** be changed at one half the normal recommended service intervals. Oil with an API classification of CA or CB **must not** be used.

The oil supplier is responsible for the quality and performance of their product.

Cummins Inc. recommends engine oil with a nominal ash content of 1 to 1.5 percent mass. Oils with higher ash content, up to 1.85 percent ash, can be used in areas where the sulfur content of the fuel is normally 1 to 1.5 percent mass. Limiting ash content is critical to the prevention of valve and piston deposit formation.

For further details and discussion of engine lubricating oils for Cummins engines, refer to service bulletin Cummins Engine Oil Recommendations, Bulletin 3810340.

New Engine Break-in Oils

Special break-in engine lubricating oils are **not** recommended for new or rebuilt Cummins engines. Use the same type of oil during the break-in as is used in normal operation.

Synthetic or partially synthetic engine oils, however, can **not** be used in a new or rebuilt engine during break-in. Use a standard petroleum-based oil for the first drain interval.

Additional information regarding lubricating oil availability throughout the world is available in the Engine Manufacturer's Association Lubricating Oils Data Book for Heavy-Duty Automotive and Industrial Engines. The data book can be ordered from the address below:

Engine Manufacturer's Association

One Illinois Center

2 N Lasalle Street

Chicago, IL U.S.A. 60601

Phone: (312) 644-6610

The oil performance classifications represented in the following chart as standard or premium have been determined through field experience and engine testing. Engine component wear and deposits are limiting factors in evaluating useful oil life. Contact a Cummins representative or oil supplier representative(s) for further assistance, if needed.

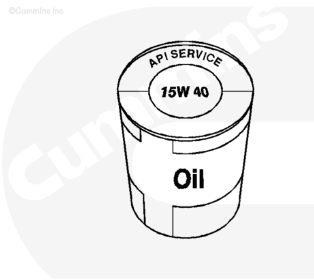
Cummins Inc. recommends the use of a high-quality 15W-40 multiviscosity, heavy-duty engine oil that meets the requirements of Cummins Engineering Standard (CES) 20071, 22072, 20075, 20076, and 20077. The oil categories CC, CD, and CE have been obsoleted by API and are **not** to be used as their specifications are no longer controlled. Engine oils that meet CES 20076 and 20077 are also considered premium oils when using the Chart Method. ACEA and API designations have **not** been assigned to CES 20076 and 20077 as of yet.

Oil Performance Classification			
API Category	**Cummins Engine Standard (CES)**	**ACEA**	**Classification of Oil Charts**
CC, CD, CE	N/A	N/A	N/A
CF-4/SG, CG-4SH	20075	E2, E3	Standard
CH-4	20072, 20071	E5	Premium
N/A	20076, 20077	N/A	Premium

ACEA = European Automobile Manufacturers Association

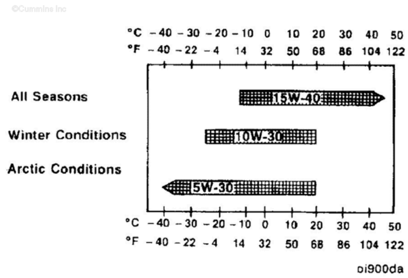
Viscosity Recommendations

The viscosity of an oil is a measure of its resistance to flow. The Society of Automotive Engineers has classified engine oils into viscosity grades. Oils that meet the low temperature (-18°C [0°F]) requirement carry a grade designation with a W suffix. Oils that meet both the low and high temperature requirements are referred to as multigrade or multiviscosity grade oils.



07a00030

Cummins Inc. has found that the use of multigrade lubricating oil improves oil consumption control and engine cranking in cold conditions while maintaining lubrication at high operating temperatures and can contribute to improved fuel consumption.



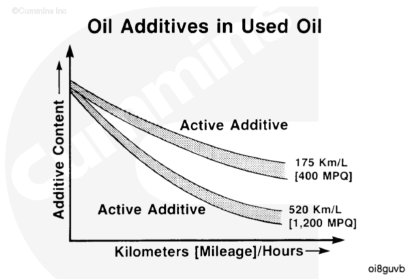
⚠ CAUTION ⚠

When single-grade oil is used, make sure the oil will be operating within the temperature ranges, as shown.

Cummins Inc. recommends the use of multigrade lubricating oils with the viscosity grades for the ambient temperatures indicated. This picture shows **only** the preferred oil grades.

Single-grade oils can be substituted for short durations until the recommended multigrade is procured. Arctic condition oils are available commercially with better low temperature properties. Consult your supplier.

The primary criterion for selecting an oil viscosity grade is the lowest temperature the oil will experience while in the engine oil sump. Bearing problems can be caused by the lack of lubrication during the cranking and start up of a cold engine when the oil being used is too viscous to flow properly. Change to a lower viscosity grade of oil as the temperature of the oil in the engine oil sump reaches the lower end of the ranges shown in the picture.



As the engine oil becomes contaminated, essential oil additives are depleted. Lubricating oils protect the engine as long as these additives are functioning properly. Progressive contamination of the oil between oil and filter change intervals is normal. The amount of contamination will vary depending on the operation of the engine, hours or miles on the oil, fuel consumed, and new oil added.

Arctic Operation

⚠ CAUTION ⚠

The use of a synthetic base oil does not justify extended oil change intervals. Extended oil change intervals can cause engine damage.

If an engine is operated in ambient temperatures consistently below -23°C [-10°F] and there are no provisions to keep the engine warm when it is **not** in operation, use a synthetic engine oil with adequate low temperature properties.

The oil supplier **must** be responsible for meeting the performance service specifications.

Refer to Section 2 for the use of the Chart Method to allow extended oil change intervals.

Last Modified: 22-Nov-2002